Match-Y Simulation Report

basic_grandfather_2_mutation_final

Pedigree based match probability results

Match probabilities

Type	Probability	Iterations	Model probabilities
Average pedigree [1]	4.4245E-05	150,000	4.3804E-05, 4.4271E-05, 4.4661E-05
Extended average pedigree [2]	1.0000E+00	1	1.0000E+00, 1.0000E+00, 1.0000E+00
Inside match [3]	• 1: 2.5005E-01 • 0: 7.4995E-01	150,000	2.4950E-01, 2.5115E-01, 2.4949E-01
Outside match [4]	0.0000E+00	0, 0, 0	0.0000E+00, 0.0000E+00, 0.0000E+00

There is a **25%** probability that at least one other (non-excluded) unknown individual in this pedigree matches the suspect haplotype. Likelihood Ratio (LR) = It is **0.333** times more likely that the suspect is the only individual in this pedigree carrying the suspect haplotype, compared to the alternative that at least one other untyped individual in the pedigree also carries it.

There is a **0%** probability that a random individual just outside this pedigree matches the suspect haplotype.

Disclaimer: The match probabilities shown above are the result of a stochastic (randomized) simulation. Due to the nature of Monte Carlo sampling, results may vary slightly between independent simulation runs, even when using the same input parameters.

[1] The average pedigree probability is the probability of the current state of the pedigree, i.e. with all typed and untyped haplotypes.
[2] The extended average pedigree probability is the probability of the current state of the extended pedigree.
[3] The inside match probability is the probability that at least one other (non-excluded) unknown individual in this pedigree matches the suspect haplotype.

[4] The outside match probability is the probability that an individual just outside this pedigree matches the suspect haplotype.

Per individual match probability

Individual	Match probability
5	13.9%
7	0.00424%
3	0.00763%
2	24.8%
6	0.00466%

Performance Metrics

- Run time pedigree probability: 0:00:45.862405
- Run time proposal distribution: 0:01:49.198284
- Run time extended pedigree: 0:00:00.000554
- Run time outside match: 0:00:00.000013
- Total run time: 0:02:35.061303
- Total needed iterations: 300,001

Simulation metadata

Date: 2025-10-06 13:50

- **Two step mutation factor:** 0.03
- **Stability window:** 25000
- **Model validity threshold:** 0.05
- **Simulation name:** basic_grandfather_2_mutation_final
- **Number of threads:** 20
- **Results path:** /media/nw_disk1/dion/repositories/MatchY/results/Basic_pedigree/basic_grandfather_2_mutation_final_20251006134729
- **User name:** dz

Pedigree information

Nodes: 7

Edges (relationships): 6

Individuals:

4: known, excluded: no

2: unknown, excluded: no

5: unknown, excluded: no

1: suspect, excluded: no

3: unknown, excluded: no

6: unknown, excluded: no

7: unknown, excluded: no

Known haplotypes

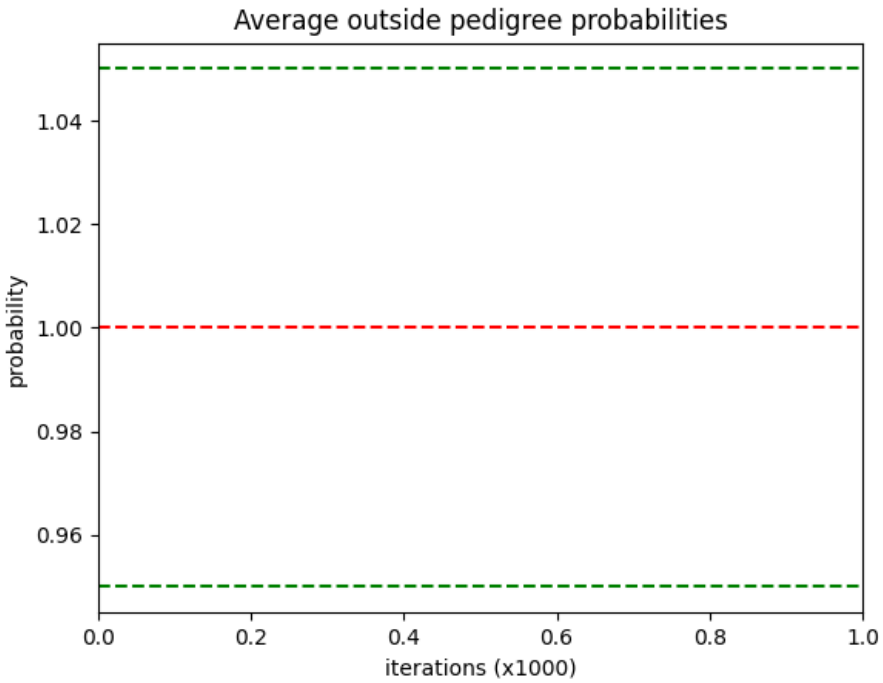
Marker	**4**	**1**
DYF1000	10, 11, 12, 13	10, 11, 12, 13
DYF1001	10, 11, 12	10, 11, 12
DYF1002	10, 11	10, 11
DYF387S1	10, 11	10, 11
DYF393S1	10	10
DYF399S1	10, 11, 12	10, 11, 12
DYF403S1a	10, 11, 12	10, 11, 12
DYF403S1b	10	10
DYF404S1	10, 11	10, 11
DYR88	10, 11	10, 11
DYS1003	10	10
DYS1005	10	10
DYS1007	10	10
DYS1010	10	10
DYS1012	10	10
DYS1013	10	10
DYS442	10	10
DYS449	11	10
DYS518	11	10
DYS526b	10	10
DYS547	10	10

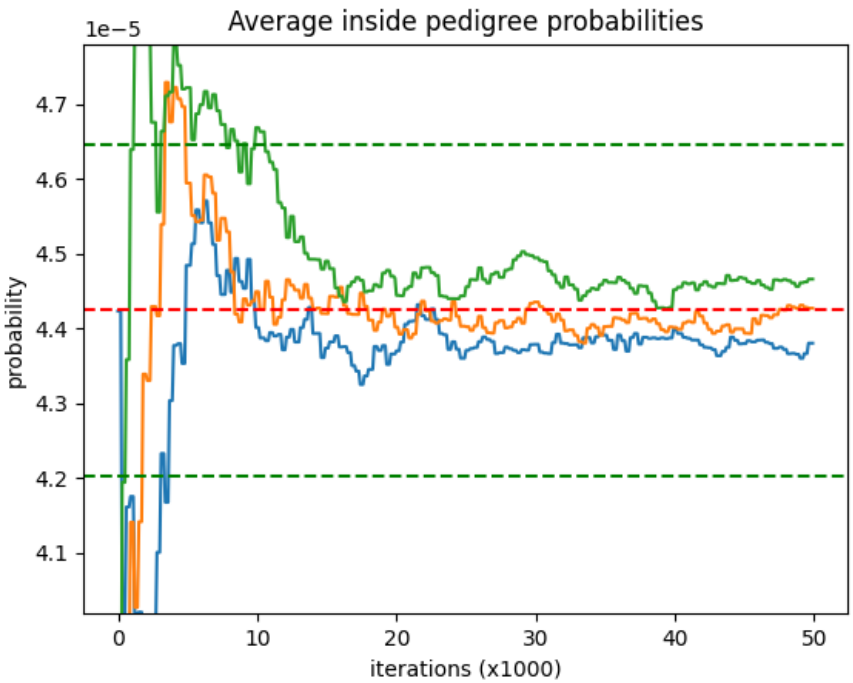
Marker	4	1
DYS570	10	10
DYS576	10	10
DYS612	10	10
DYS626	10	10
DYS627	10	10
DYS711	10	10
DYS712	10	10
DYS713	10	10
DYS724	10, 11	10, 11

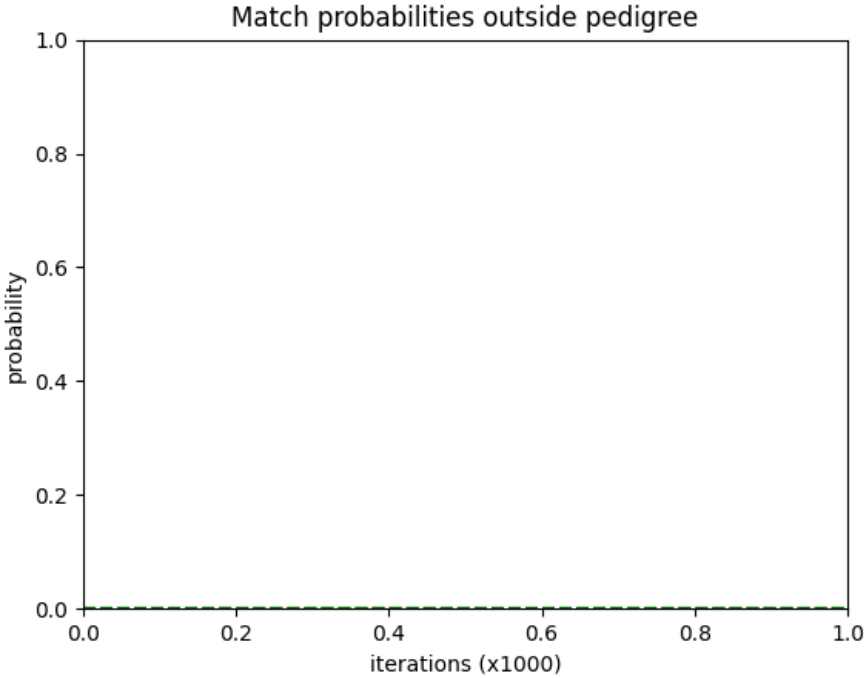
Marker set (marker: mutation rate)

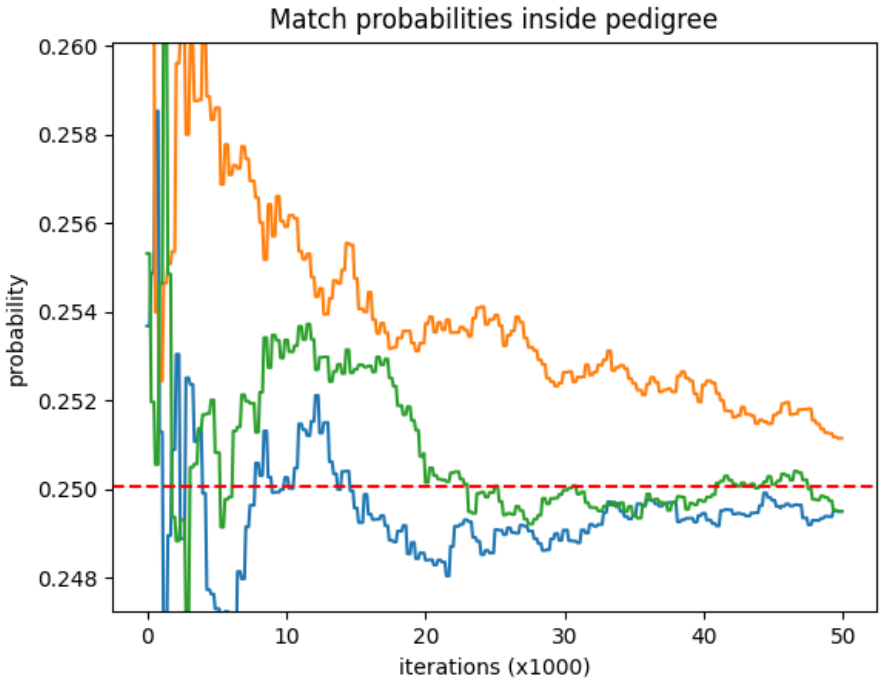
- DYF1000: 0.035880708
- DYF1001: 0.048041045
- DYF1002: 0.016775396
- DYF387S1: 0.010224215
- DYF393S1: 0.007136485
- DYF399S1: 0.062834749
- DYF403S1a: 0.027253957
- DYF403S1b: 0.018151354
- DYF404S1: 0.012512031
- DYR88: 0.026328162
- DYS1003: 0.012581547
- DYS1005: 0.009785648
- DYS1007: 0.017241379
- DYS1010: 0.013999067
- DYS1012: 0.01584343
- DYS1013: 0.010782935
- DYS442: 0.000838379
- DYS449: 0.011216776
- DYS518: 0.013288448
- DYS526b: 0.02464744
- DYS547: 0.014683544
- DYS570: 0.00827857
- DYS576: 0.012674578
- DYS612: 0.016267943
- DYS626: 0.008596713
- DYS627: 0.014489091
- DYS711: 0.026561044
- DYS712: 0.031098546
- DYS713: 0.013859275
- DYS724: 0.048018648

Model validation plots



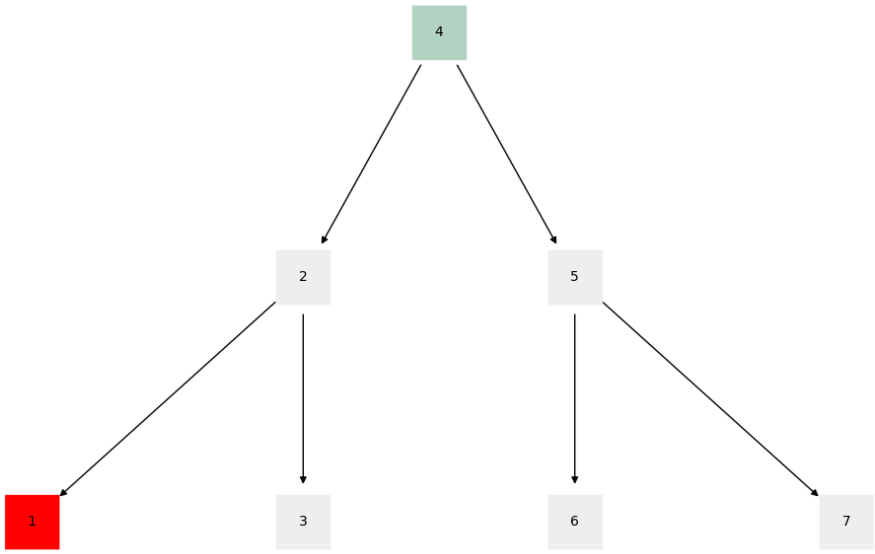



Match probabilities outside pedigree
probability
iterations (x1000)
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0

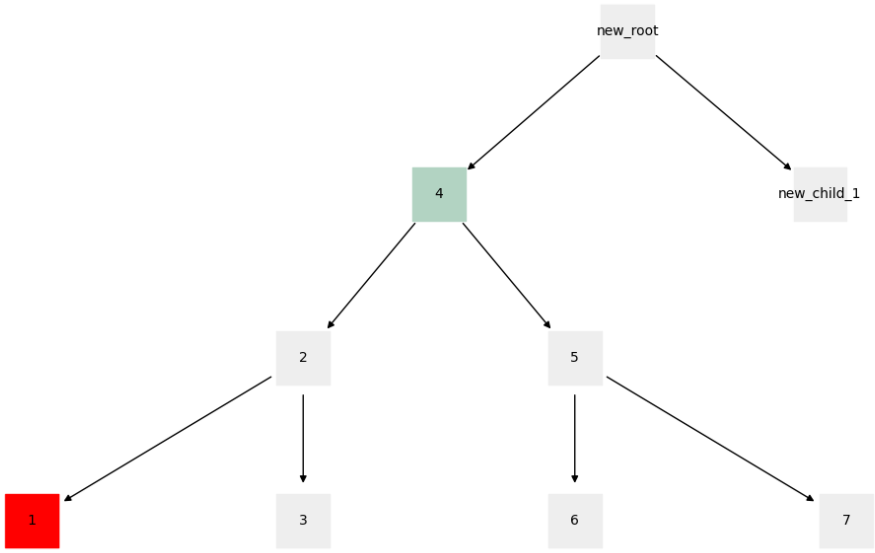

Match probabilities inside pedigree
probability
iterations (x1000)
0.260
0.258
0.256
0.254
0.252
0.250
0.248
0
10
20
30
40
50

Pedigree configurations

Original Pedigree



Extended Pedigree



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